Muhammad Abdullah Zahid

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Research Interests

Mixed Reality Interfaces for Fine Robotic Telemanipulation. How the operator's visual representation of the robot and its workspace shapes fine manipulation, and whether spatially registered mixed reality helps operators not only perform better but also acquire skills faster.

Skill Transfer from Virtual Training to Physical Execution. Whether a single interface shared by virtual training and real teleoperation lets skills carry over without readaptation, and how digital twins can keep that interface faithful to a changing remote workspace in real time.

Wearable Haptics for Telemanipulation and XR. Which manipulation events are worth conveying through touch, and how lightweight wearables, vibrotactile or passive kinesthetic, can render them without tethering the operator to a grounded device.

Autonomous Robot Control. Robust semantic object navigation for mobile robots in dynamic, uncertain environments.

Education

Kyoto University of Advanced Science (KUAS), Kyoto, Japan — Sep. 2027 – Sep. 2029
Master of Engineering | *Admitted; enrolling Sep. 2027 (two-year program)*
Graduate School of Engineering | *Division of Mechanical and Electrical Systems Engineering*

- **Research Supervisor** Dr. Sajid Nisar, Novel Intelligent Systems & Advanced Robotics Laboratory
- **Planned Research** Bridging Training and Execution: A Unified Mixed Reality Interface for Robotic Telemanipulation

Kyoto University of Advanced Science (KUAS), Kyoto, Japan — Sep. 2023 – Sep. 2027
Bachelor of Engineering | *Expected graduation: Sep. 2027*
Faculty of Engineering | *Department of Mechanical and Electrical Systems Engineering*

- **Research Supervisor** Dr. Sajid Nisar, Novel Intelligent Systems & Advanced Robotics Laboratory (Mar. 2025 – Present)

Research Experience

NISAR Laboratory, KUAS, Undergraduate Researcher — Mar. 2025 – Present
Research Supervisor: Dr. Sajid Nisar | *Novel Intelligent Systems & Advanced Robotics Laboratory*

- **HaptLinX: Haptic-Enabled MR Telemanipulation Training (2025) [C4].** First prototype: a digital twin of a 6-DoF Kinova Gen3 Lite, controlled in MR on a Meta Quest 3S through hand tracking and mirrored in real time by the physical arm, with bHaptics glove and suit cues for contact and alignment. Selected for the IEEE World Haptics 2025 Student Innovation Challenge.
- **MR Training Interface for Fine Telemanipulation (Lab Project, 2026 – Present) [C1].** Built the training module of the unified MR interface and evaluated it in a twelve-participant 2×2 within-subject study against a 2D multi-view display under the same control mapping. MR raised performance (5.19 vs. 1.19 successful placements/trial) and learning rate (+0.479 placements/trial); vibrotactile cues cut self-collisions by 0.71 per trial but did not accelerate learning.
- **Unified MR Interface from Training to Execution (In Progress; Master's Research from Sep. 2027).** Extending the same MR interface and hand-based control from virtual training to teleoperation of the physical robot through its digital twin, with the robot workspace synchronised through RGB-D object reconstruction and pose tracking (NVIDIA FoundationPose), to measure how skills transfer to real execution.

- **Escapement-Like Kinesthetic Rendering for a Virtual Piano [C2].** Designed a displacement-dependent controller that renders a piano key's let-off through a passive wearable tendon-resistance device with no key surface. Force-proxy measurements over five device fittings showed a consistent drop at let-off, then recovery; ten participants detected the release with 95% median accuracy and located it within the keystroke with 80%.
- **Wearable vs. World-Grounded Haptics for Space Teleoperation [C3].** Contributed to a VR comparison of a world-grounded (sigma.7) and a wearable (Dexmo) interface, with and without force feedback, on a fine-manipulation task modelled on space robotics.
- **Uncertainty-Aware Semantic Object Navigation (2026 – Present).** Developing an ObjectNav framework that combines object memory, detection confidence, and spatial uncertainty with active exploration, in NVIDIA Isaac Sim.

Publications

Peer-Reviewed Conference Papers

[C1] Skill Acquisition in Robotic Teleoperation: Effects of Mixed Reality and Vibrotactile Feedback
M. A. Zahid, R. M. Suarez Flores, S. Nisar
2026 9th International Conference on Robotic Systems and Applications (ICRSA 2026). IEEE Xplore indexing in progress.

[C2] Escapement-Like Kinesthetic Rendering with a Passive Wearable Device for a Virtual Piano
M. A. Zahid, S. Shukla, R. M. Suarez Flores, S. Nisar
2026 9th International Conference on Robotic Systems and Applications (ICRSA 2026). IEEE Xplore indexing in progress.

[C3] Wearable vs. World-Grounded Haptics for Space Teleoperation: A VR-Based Comparative Analysis of Kinesthetic Feedback for Fine Telemanipulation
K. H. Joshi, R. M. Suarez Flores, **M. A. Zahid**, S. Nisar
31st International Symposium on Artificial Life and Robotics 2026 (AROB 2026).

Non-Peer-Reviewed Papers

[C4] Developing a Haptic-Enabled Mixed-Reality System for Fine Robotic Telemanipulation Training
M. A. Zahid, R. M. Suarez Flores, K. K C, K. Meitani, A. Majumder, S. Nisar
43rd Annual Conference of the Robotics Society of Japan (RSJ2025).

Presentations

Oral ICRSA 2026, Huizhou, China — [C1] and [C2] (Oct. 30 – Nov. 1, 2026).
Oral RSJ2025, International Session — Human and Robot, Tokyo, Japan — [C4] (Sep. 2025).
Poster IEEE World Haptics Conference 2025, Student Innovation Challenge — Mixed Reality, Suwon, South Korea — *HaptLinX* (Jul. 2025).

Teaching Experience

Introduction to Robotic Manipulators, Teaching Assistant, KUAS — Sep. 2026 – Present
Algorithmic Thinking and Programming with Python, Teaching Assistant, KUAS — Apr. – Jul. 2026

Projects and Leadership

LLM-Driven Natural Language Robot Control, Team Leader, with AsiaQuest Co., Ltd. — 2025

- Led a five-member team building a mobile robot that executes spoken multi-step commands locally (Whisper, Mistral 7B, Grounded SAM 2 on RGB-D, ROS 2 SLAM and Nav2); worked on the language processing, vision, and navigation pipelines.

Kyorabo, Co-Founder 2026 – Present

- Research-collaboration matching app built by a team of five (React Native, Supabase), live on iOS and the web; reached 80 researchers from 45 institutions in 18 countries in its first week. [Website] [Devpost]

AirKeys, KUAS Cornerstone Project 2025 – 2026

- Self-initiated virtual piano: a VR version with hand tracking (2025), extended to MR with AI features and haptic feedback (2026); the origin of the haptic rendering work in [C2].

Mobile Robots, Mechatronics Laboratory 2025 – 2026

- *ELIBot*: four-wheeled obstacle-course robot teleoperated through ROS 2 on a Raspberry Pi 4, with QR-code and traffic-light detection and STM32 motor control [GitHub]; *self-balancing line follower* with staged LQR control [GitHub].

Memberships and Service

Memberships IEEE, Student Member; Robotics Society of Japan, Student Member (2025 – Present).
Outreach Featured Speaker, "Study in Japan: Fully Funded Undergraduate Scholarships," Pakistan Student Association Japan (PSAJ), online (Mar. 2026).

Skills

XR and Haptics Unity (C#), Meta Quest 3S, digital twins, hand tracking, inverse kinematics, bHaptics TactSuit Pro / TactGlove DK2, sigma.7, Dexmo.
Robotics Kinova Gen3 Lite, teleoperation, ROS 2, NVIDIA Isaac Sim, computer vision, mobile robot navigation, embedded control (Arduino, STM32).
Research Methods Within-subject experiment design, user studies, statistical analysis, LaTeX.
Programming Python, C/C++; Linux, Git.
Applied AI Integrating pretrained models: Grounded SAM 2, SAM 3D, FoundationPose, Whisper, Mistral 7B.
Languages English (IELTS 7.5), Japanese (JLPT N4).
Certifications JSPS Research Ethics, eL CoRE (2026) [Certificate]; MITx 6.00.1x, Computer Science and Python (2024) [Certificate].